STATS 402 - Interdisciplinary Data Analysis

Short-Horizon Temporal Optical Flow with Physics-Informed Consistency

Milestone Report: Stage 1

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Abstract

Optical flow estimation is a fundamental problem in computer vision that aims to predict dense pixel-wise motion between consecutive frames. It serves as a critical component in practical applications such as autonomous driving, object tracking and scene understanding. While real-world motion is inherently temporal and structured, classical optical flow methods are typically formulated on pairs of frames and treat pixels largely independently. As a result, these methods are limited in the ability to capture temporal coherence and structured motion dynamics that are implied across multiple timesteps. This project aims to extend this classical two-frame formulation toward a short-horizon temporal modeling framework, specifically by treating motion as an evolving spatial field. We propose to learn motion representations from sequences of frames, enabling the model to capture temporal coherence and evolving motion patterns beyond pairwise correspondence. To achieve this, we introduce an operator-based temporal module that models the evolution of latent motion features across time. Inspired by recent advances in operator learning and spectral convolution, this module is designed to capture non-local spatial dependencies and structured temporal dynamics in motion fields. Rather than explicitly enforcing governing partial differential equations, the proposed approach implicitly learns motion evolution patterns from data while retaining lightweight physics-informed regularization, including temporal consistency and spatial smoothness constraints. These constraints act as stabilizing inductive biases, while the operator-based module provides a stronger representation of motion dynamics than conventional frame-to-frame consistency losses.

By integrating pairwise optical flow estimation with learned spatiotemporal evolution, this framework aims to improve temporal coherence, stability, and physical plausibility of predicted motion fields. The proposed approach bridges classical optical flow formulations with modern operator-based learning, offering a lightweight yet structured pathway toward multi-frame motion understanding in dynamic scenes.

1. Project Rationale

1.1 Dataset and Characteristics

Optical flow and scene flow estimation aim to recover dense motion fields from image sequences, serving as a fundamental component in tasks such as autonomous driving, 3D reconstruction, and scene understanding. In this project, we intend to utilize the KITTI Scene Flow dataset, which represents real-world dynamic environments characterized by multiple independently moving objects, camera motion, and complex geometric structures. However, from the theoretical perspective, motion estimation is inherently ill-posed, where only the normal component of motion can be directly inferred from image measurements, with the full motion field requiring additional constraints, regularizations, and assumptions (Vedula et al., 2005). These limitations persist when extending from optical flow in 2D to scene flow in 3D. Therefore, effective methods must incorporate both data-driven learning, and structural priors to resolve the inherent ambiguities.

1.2 Related Works and Development Trends

A. Classical methods

Early approaches formulate optical and scene flow estimation as variational optimization problems, combining data fidelity and regularization terms. These methods rely on assumptions such as geometry, spatial smoothness, piecewise rigidity and temporal consistency. For example, multi-frame methods exploit temporal coherence to improve robustness and accuracy (Vogel et al., 2014). However, such approaches rely heavily on artificial priors and may struggle with complex real-world scenarios.

B. Deep learning, data-driven methods

More recent advances have shifted towards deep learning frameworks, including Convolutional Neural Network (CNN) and Transformer-based models. These methods learn motion representations directly from data and could achieve strong performance on benchmark datasets. For example, transformer-based approaches are able to capture global spatial dependencies and temporal associations via their attention mechanism, improving robustness under occlusion and large motion (Lu et al., 2024). Similarly, task-specific improvements such as key-point-aware flow estimation highlight the importance of incorporating structural cues in learning-based models (Yi et al., 2024). Nevertheless, these methods primarily treat flow estimation as a spatiotemporal correspondence learning problem, with limited consideration of the underlying physical or dynamic structure of actual motions.

C. Operator-based scientific machine learning

In parallel with recent advancements in deep learning, scientific machine learning has introduced neural operator frameworks, such as the Fourier Neural Operator (FNO), which enables learning mappings between function spaces and approximate solution operators of differential equations (Li et al., 2021). Such methods provide strong inductive bias for both discrete and continuous systems, enable generalization across discretization, and efficient approximation of complex dynamic systems. Subsequent works have extended this research to better capture multiscale structure and improve computational efficiency. For instance, U-FNO augments spectral operators with U-Net-style local convolutional pathways to enhance high-frequency detail modeling (Wen et al., 2022), while U-NO generalizes this idea through a U-shaped neural operator architecture that supports deeper and more memory-efficient learning on high-resolution spatiotemporal domains (Rahman et al., 2023). This emerging paradigm of operator-based frameworks reveals the potential of modeling physical processes as function approximators, rather than data-driven discrete mappings, and suggests a promising direction for incorporating structured inductive biases into motion learning problems.

1.3 Commonly Used Methods and Existing Limitations

Across existing approaches, three major categories can be identified:

1. Variational methods,

These methods adhere to realistic constraints that the physical environment imposes on the optics of images. Therefore, although these methods provide strong theoretical background and are generally interpretable, they are limited in expressiveness and scalability.

2. Deep learning / Data-driven methods,

These methods are based on deep learning with substantial data input, which to an extent overcomes the more traditional paradigm and induces stronger representation and learning capacity. However, these methods often introduce pure data approximators that conduct discretized data mappings, and are often expelled from physical consistency. Frameworks belonging to this category also rely heavily on the accessibility of data.

3. Hybrid / Self-supervised methods,

Due to the constraint of data availability, most works on multi-frame flow estimation relies on self-supervision since ground truth is mostly unavailable. This category of methods synthesize data, which reduces dependence on labeled data (Neumann et al., 2025). However, similar to general deep learning methods, such indirect supervision is to some extent unstable under real-world scenarios.

1.4 Our Motivation

Based on the above analysis, the core motivation of this project is therefore to bridge the gap in existing methods that are less able to fully exploit the structured and dynamical nature of motion, which were primarily treated as static matching problems, rather than a temporal evolution process. Furthermore, we also intend to extend the existing data-driven vision modeling approaches with a physics-inspired method that encourages the framework to learn real-world adherence. Therefore, it is necessary to explore new frameworks that better align with the intrinsic properties of motion in real-world scenes.

1.5 Application Prospects

Accurate motion estimation plays a critical role in autonomous driving, robotics and general 3D vision tasks. The estimations are able to provide substantial information for obstacle detection, trajectory prediction, scene understanding, motion planning,

etc. Given the increasing demand for reliability in safety-critical systems, the need for methods that are not only accurate but also robust, interpretable and physically consistent has also been rising.

2. Research Content, Objectives and Limitations

2.1 Research Content

1. Contents

This project aims to research the integration of operator-based deep learning frameworks into optical and scene flow estimation. Specifically, we extend the traditional pairwise and single flow estimation toward short-horizon temporal prediction modeling motion as an evolving dynamical process over multiple frames, without relying on the computational based attention mechanism. The proposed framework combines a data-driven image encoder with an operator-learning module (e.g., FNO-like components) to capture global spatial and temporal motion fields. More broadly, this project may serve as a research on combining FNO-like modules in visual representation learning, which may have wider implications for modeling underlying physical processes from multi-channel spatiotemporal data.

2. Timeline

Week 1-2

Finalize project scope and formulation;

Validate dataset setup and preprocessing pipeline;

Design overall model architecture.

Week 3-4

Confirm framework pipeline;

Implement baseline image encoder and flow estimation module;

Develop initial operator module;

Integrate components into a unified pipeline.

Week 5-6

Finish operator implementation and refinement;

Conduct full model training;

Evaluate model performance and analysis.

2.2 Research Objective

This project primarily aims to produce a functional deep learning framework that is capable of producing temporally consistent short-horizon optical and scene flow estimation given sequences of input image frames. The project also aims to provide implications on whether operator-based function approximators may advance physics-informed learning without observed governing equations, bringing the classical optical flow problems in a contemporary deep learning and dynamic optimization approach.

2.3 Limitations

A key limitation of this project lies in the restricted supervision provided by the selected dataset (KITTI). The dataset primarily offers ground truth optical flow for individual frame pairs, while longer sequences (e.g., 20-frame clips) contain only raw RGB images without corresponding motion flows. This constraint requires the use of self-supervised learning strategies to model temporal dynamics across multiple frames. However, such approaches introduce challenges, including underdetermined temporal learning, error accumulation and limited physical interpretability.

Availability and access to ground truth motion fields across consecutive frames would significantly improve training stability, enable stronger evaluation of temporal coherence, and enhance the interpretability of the learned operator as a global function approximator of motion dynamics.

3. Proposed Approach

3.1 Methods and Technical Routes

This project focuses on developing a spatiotemporal motion learning framework that moves beyond traditional correspondence-based formulation of optical and scene

flows. The key idea is to treat motion not as static mapping between frames, but as continuous dynamic processes that are evolving over space and time. Thus, the research content of this project should include

1. Spatiotemporal representation of motion fields

We first reformulate the classic optical and scene flows as a spatiotemporal field, for the 2-dimensional optical flow:

$$u(x,t)$$

for the 3-dimensional scene flow:

$$v(x,t)$$

where $x = x(t)$ is the 3-dimensional path of a point, and t being time. Unlike traditional 2-frame estimation, this formulation explicitly indexes temporal evolution across multiple frames and considers the consistency of motion trajectories. Specifically, for images I_t at time t, we encode

$$\widehat{F_{t-1\to t}} = E_\phi(I_{t-1}, I_t),$$

$$\widehat{F_{t\to t+1}} = E_\phi(I_t, I_{t+1}),$$

$$\widehat{F_{t+1\to t+2}} = E_\phi(I_{t+1}, I_{t+2})$$

or more generally,

$$\{\widehat{F_{t-i\to t-i+1}}\}_{i=1}^{k},$$

where $\widehat{F_{t\to t+1}}$ represents the discretized motion field $u(x,t)$ between consecutive image frames. We further put these into a new encoder E_θ, where

$$Z_t = E_\theta\left(I_{t-1:t+2}, \widehat{F_{t-1\to t}}, \widehat{F_{t\to t+1}}, \widehat{F_{t+1\to t+2}}\right)$$

where Z_t becomes the spatiotemporal feature embedding that captures visual content from images, motion evolution from flows, while ensuring short-horizon temporal consistency. Figure 3.1.1 provides a visualization of this formulation.

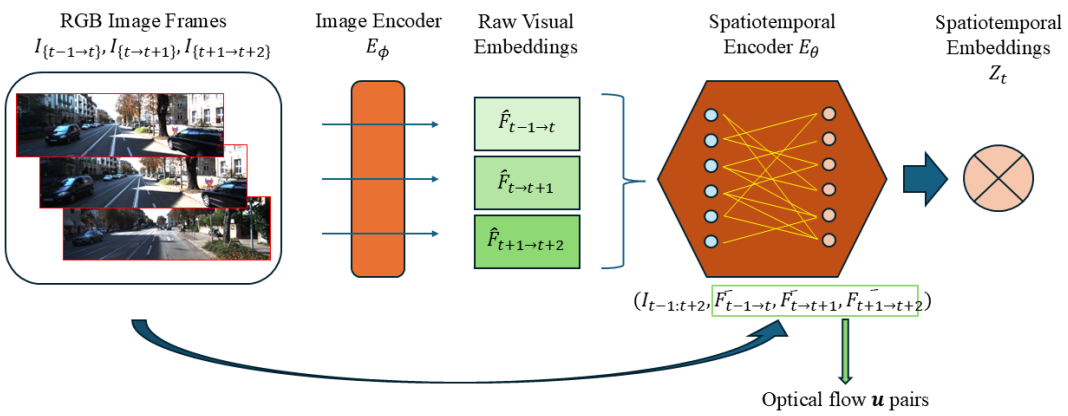

Figure 3.1.1: Spatiotemporal representation of motion fields

2. Learning motion evolution

Building on the spatiotemporal formulation of motion fields, we shift from estimating independent pairwise frame displacements toward modeling the temporal evolution of motions. In classical optical flow, pixel motion is locally constrained through differential relationships such as:

$$\frac{du}{dt}$$

which describes the dependent relationship between change of motion and time. However, as shown by (Vedula et al., 2005), such constraints are inherently underdetermined, with only the normal part solvable, leaving the orthogonal component unsolvable. Thus, it requires additional structure or regularization to recover the full motion fields.

Instead of treating this limitation as a constraint problem, we would instead interpret this as a dynamical modeling problem. Specifically, we assume that the motion fields evolving according to some unknown governing equation:

$$\frac{\partial u}{\partial t} = \mathcal{F}(u, I, x, t)$$

$$\frac{\partial v}{\partial t} = \mathcal{F}(v, I, x, t)$$

where $\mathcal{F}$ represents an implicit dynamical operator that governs the evolution in space and time. This formulation shifts the project from reconstructing a pairwise correspondence estimation for learning how motion propagates and transforms in time, enabling the model to capture temporal coherence and structured motion patterns across multiple frames.

3. Modeling operator-based framework for motion dynamics

To approximate the unknown evolution operator $\mathcal{F}$, we adopt an operator-based learning framework inspired by neural operators. Instead of learning discrete data point mappings, the model learns a mapping between functions:

$$\mathcal{T}_\theta: Z_t \rightarrow Z_{t+1}$$

where $\mathcal{T}_\theta$ is a learned temporal evolution operator in the latent space. Practically, the method would be implemented using spectral or Fourier-based convolutional units, which enable efficient modeling of continuous, global spatial dependencies, and nonlocal interactions within motion fields. Meanwhile, it is worth noting that vanilla FNO frameworks are explicitly biased toward lower frequency signals, which truncates higher frequency ones that are associated with abrupt jumps in PDE solutions (Li et al., 2021). However, as sharp-edged objects and artificial occlusions exist in real-world optical and scene flows, the model can benefit from multi-resolution processing via down sampling, which effectively re-parameterizes fine-scale variations into lower-frequency representations that can be captured by spectral convolutions, as in U-NO structures (Rahman et al., 2023).

4. Integration with data-driven learning

While the operator module models motion evolution, it is integrated within a data-driven learning framework to ensure practical effectiveness. A base optical flow backbone first extracts pairwise motion estimates and image features, which are then encoded into a latent spatiotemporal representation:

$$Z_t = E_\theta\left(I_{t-k:t+k}, u_{frame\ pairs}\right)$$

The operator module then evolves this representation over time:

$$Z_{t+1} = \mathcal{T}_\theta(Z_t)$$

and a decoder maps the representation back to refined flow predictions:

$$\widehat{u_{t\to t+1}} = D_{\psi}(Z_{t+1})$$

In order to stabilize learning and ensure physically meaningful motion, lightweight regularization in terms of spatial smoothness, temporal consistency will be incorporated. These will act as soft constraints that complement the operator's ability to learn structured motion dynamics from data. Figure 3.1.2 demonstrates this workflow.

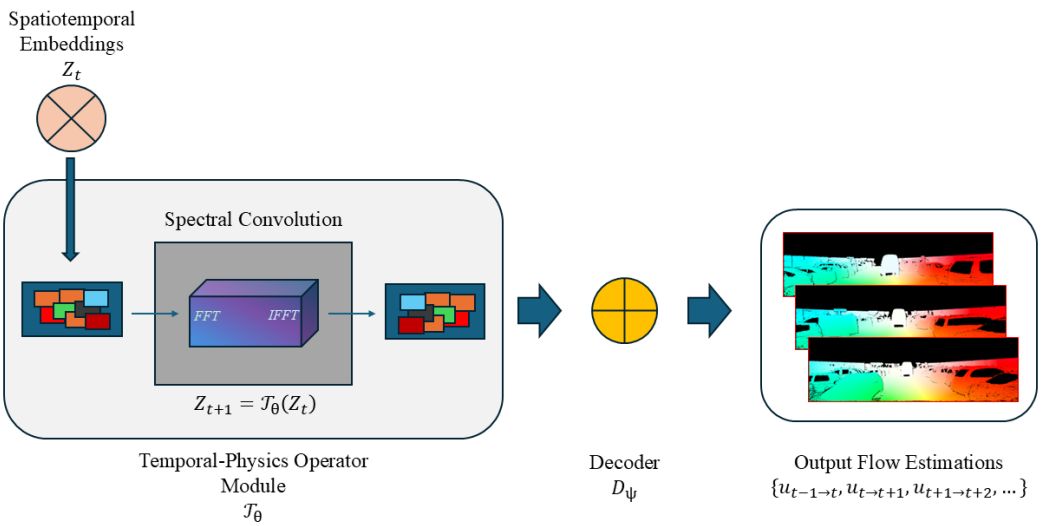

Figure 3.1.2: Operator-based spatiotemporal motion evolution learning

Overall, the framework takes as input a short sequence of consecutive RGB frames (e.g. 3-5 frames), and outputs temporally consistent optical (or scene) flow predictions between frames, and can be extended to produce refined multi-step motion fields or trajectory representations.

3.2 Architecture Design and Model Selection

To ensure both computational efficiency and effective spatiotemporal modeling, the proposed framework adopts a fully convolutional architecture without relying on attention mechanisms. The overall design follows a modular structure, consisting of four main components: pairwise flow encoder, joint image–flow encoder, operator-based temporal module, and decoder. The architectural choices are motivated by both performance considerations and the specific requirements of motion modeling.

1. Pairwise Flow Encoder

The pairwise flow encoder is designed using a convolutional neural network (CNN) combined with a correlation (cost volume) layer. Given two consecutive frames I_t and I_{t+1}, shared CNN backbones extract feature representations, and a correlation operation computes dense correspondences between feature maps.

This design is inspired by classical optical flow architectures such as PWC-Net (Sun et al., 2018) and RAFT (Teed & Deng, 2020), where explicit correspondence modeling is critical for accurate motion estimation. Compared to transformer-based matching mechanisms, correlation layers provide a more computationally efficient alternative while preserving the ability to capture pixel-wise relationships.

The use of a CNN-based encoder ensures efficient feature extraction, compatibility with dense prediction tasks and avoidance of heavy attention computation. Therefore, this module provides a reliable and lightweight estimation of pairwise motion fields $\{\widehat{F_{t-i\to t-i+1}}\}_{i=1}^{k}$, which serves as a fundamental input for subsequent temporal modeling.

2. Joint Image–Embedding Encoder

To integrate appearance and motion information, we intend to adopt a two-stream CNN architecture. The structure consists of two branches: one branch processes RGB image sequences, while the other processes the image embeddings computed in step 1. The input features are concatenated and further processed through a temporal fusion module. Instead of using attention-based temporal modeling, we adopt frame-stacked 2D convolution. This design enables the model to capture short-horizon temporal dependencies and especially maintains computational efficiency.

The motivation for this design is threefold: separating image and flow processing improves representation learning; convolutional fusion is significantly more efficient than attention and short-horizon temporal modeling aligns with the project objective.

As a result, the encoder produces a latent representation: $Z_t = E_\theta(I_{t-k:t+k}, \hat{F})$, which encodes both spatial structure and temporal motion patterns.

3. Operator-based Temporal Module

The temporal evolution of motion is modeled using a U-shaped Neural Operator (U-NO) (Rahman et al., 2023), extending the Fourier Neural Operator (FNO) framework (Li et al., 2021). Unlike conventional neural networks that learn point-wise mappings, neural operators learn mappings between functions, which is the evolution:

$$\mathcal{T}_\theta: Z_t \rightarrow Z_{t+1}$$

in our scenario.

The U-NO architecture combines spectral convolution layers for global spatial interaction and U-Net-like multi-resolution structure for local detail preservation, which is particularly suitable for motion modeling since motion fields exhibit long-range spatial dependencies. Therefore, global interactions are critical for dynamic systems and multi-scale representation, which theoretically improves robustness under the prevalent occlusion and discontinuity in the raw input data. It is also notable that U-NO provides a more efficient structure in terms of computational requirements, relieving training pressure.

4. Decoding Stage

The decoder is implemented using a PWC-style up-sampling architecture with a context refinement module (Sun et al., 2018). Starting from the latent representation Z_{t+1}, the decoder progressively upsamples features to reconstruct dense motion fields. Skip connections and multi-scale refinement are incorporated to preserve spatial detail, while a final context network with dilated convolutions improves local consistency and boundary accuracy.

This design is motivated by the effectiveness of multi-scale decoding in optical flow (Sun et al., 2018); the need for accurate dense prediction and the requirement for computational efficiency.

The decoder produces the final motion estimation: $\widehat{F_{t\rightarrow t+1}} = D_\psi(Z_{t+1})$. At this stage the goal of our framework is achieved.

5. Summary of Design Choices

The overall architecture is designed to balance efficiency, interpretability, and modeling capability:

- CNN-based modules ensure efficient local feature extraction

- Correlation layers provide explicit motion correspondence
- Neural Operator captures global temporal dynamics
- PWC-style decoder reconstructs high-quality dense flow

Importantly, the entire framework avoids attention mechanisms, making it suitable for limited computational resources while still enabling structured and physically meaningful motion modeling.

3.3 Experiment and Evaluation

To evaluate the effectiveness of the proposed approach, both qualitative and quantitative analyses will be conducted. We will qualitatively visualize flow fields and motion trajectories to examine spatial and temporal consistency, with particular attention to occlusions, motion boundaries, and large displacements. Quantitatively, we will test on KITTI standard benchmarks with metrics including Endpoint Error (EPE). Comparative experiments against classical and deep learning baselines will be done based on the researchers' report in papers to assess improvements in accuracy, robustness and generalization.

4. Features, Innovations and Expected Results

The main feature of this project lies in the reformulation of optical and scene flows estimation from a static, pairwise correspondence problem to a spatiotemporal dynamical modeling problem. Unlike conventional approaches that either rely on derived regularization or purely data-driven matching, this work introduces an operator-based perspective to model how motion evolves over time. By integrating ideas from neural operators with modern deep learning frameworks, the proposed method provides a structured inductive bias that captures continuous and global motion dynamics while maintaining the flexibility of data-driven learning. This combination represents a conceptual innovation that bridges the gap between physics-inspired modeling and vision-based approaches. Furthermore, the project introduces a multi-frame temporal learning method with explicit modeling of motion evolution, rather than implicit feature aggregation. Based on formulated differential relationships between points in images and time, we are able to use spectral or Fourier-based

operator modules that further enables efficient modeling of global spatial relationships. These design choices collectively aim to improve temporal coherence, robustness to occlusions and noise, and generalization abilities.

The expected results include improved accuracy and stability of motion estimation on benchmark datasets. Beyond quantitative improvements, this framework is also expected to produce physically plausible motion field estimations. More broadly, this project could provide a new perspective on motion modeling in computer vision, offering a foundation for potential future work that integrates dynamical systems and operator learning into visual representation learning.

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